$$R_1' = 3R_2$$

$$R_1' = 3 \text{ cm}$$

$$R_2 = 1 \text{ cm}$$

$$x^2 + (y-1)^2 = 1 \qquad (20)$$

$$\frac{\overline{O_2F}}{\overline{PF}} = \frac{\overline{PH}}{\overline{HQ}} \qquad (21)$$

$$\frac{x_P}{(1-y_P)} = \frac{(3-y_P)}{(3-x_P)} \qquad (22)$$

$$x_P^2 + y_P^2 - 4y_P = 3x_P - 3$$

$$x_P^2 + (y_P-1)^2 - 2y_P - 1 = 3x_P - 3$$

$$-2y_P = 3x_P - 3$$

$$y_P = \frac{3}{2}(1-x_P) \qquad (23)$$

$$y_P = 0.354 \text{ cm} \qquad (24)$$